Shaping Emergent Behavior in Multi-Agent PPO: A Five-Stage Ablation Study on Knights-Archers-Zombies

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Abstract

We present a structured five-stage ablation study of reward shaping and architectural design in a shared-parameter Multi-Agent Proximal Policy Optimisation (MAPPO) system, evaluated on the Knights-Archers-Zombies (KAZ) cooperative environment from PettingZoo. Starting from a bare kill-reward baseline (**G1**: *Greedy Soldier*), we progressively stack ammo restrictions (**G2**), stamina penalties (**G3**), 60/40 team-health rewards with entity-level self-attention (**G4**), and finally Gaussian observation fog with GRU temporal memory (**G5**: *Fire Discipline*). Our central finding is the diagnosis and resolution of *deterministic passivity*—a policy collapse phenomenon where agents trained with strong penalties learn to avoid all action under a greedy (argmax) policy. **G5** achieves a deterministic attack rate of 69.9%, compared to ≈0% in **G3** and **G4**, while simultaneously exhibiting cooperative preservation. All code, checkpoints, and visualisations are available in the project repository.

1. Introduction

Cooperative multi-agent reinforcement learning (MARL) presents unique challenges beyond single-agent RL: agents must coordinate emergent behaviour without explicit communication, manage shared resources, and remain robust to partial observability. Reward shaping is a powerful but double-edged tool—it can guide agents toward desired behaviours while inadvertently suppressing beneficial exploration or inducing unintended policy collapse [5].

This work asks a focused question: *how does the progressive addition of environmental constraints and architectural upgrades shape the emergent behaviour of MAPPO agents?* We operationalise this through a five-game ablation where each game adds exactly one new element, enabling clean attribution of behavioural changes.

The Knights-Archers-Zombies environment [7] provides an ideal testbed: four cooperative agents (2 knights, 2 archers) must eliminate zombies while managing team health. The combination of close-range melee and ranged combat creates natural specialisation pressure, while the scoring function based on kills introduces a tension between individual efficiency and team preservation.

Our contributions are:

1. A five-stage ablation demonstrating how constraint stacking drives *behavioural collapse* then *recovery* in MAPPO agents.
2. Diagnosis of *deterministic passivity*—a novel failure mode where the stochastic policy learns to be aggressive but the deterministic policy is passive—and its resolution via GRU recurrence.

3. An architectural progression from MLP to entity-attention to attention+GRU, with empirical evidence that temporal memory is the key ingredient for resolving passivity under partial observability.
4. Gradient-based saliency and attention weight visualisations comparing where **G1** (MLP) and **G5** (Attn+GRU) focus.

2. Background & Related Work

MAPPO. Yu et al. [8] showed that Proximal Policy Optimisation [6] with shared parameters achieves competitive performance in cooperative MARL without the need for centralised critics or communication channels. We follow their shared-parameter formulation where all agents share a single actor-critic network.

Reward shaping. Ng et al. [5] prove that potential-based reward shaping preserves the optimal policy under certain conditions. However, non-potential-based shaping (e.g. stamina penalties) can distort the optimal policy. Our ablation deliberately explores this distortion as a feature, demonstrating how over-constraining without architectural support induces passivity.

Attention in MARL. Iqbal and Sha [3] introduced actor-attention-critic, using multi-head attention to dynamically weight agent contributions. We apply entity-level self-attention within each agent's observation (over the 27 entity slots) rather than across agents, providing permutation-invariant feature extraction.

Recurrent MARL. Hausknecht and Stone [2] demonstrated that GRU hidden states allow policies to integrate information across time steps in partially observable environments. We add a GRUCell after the attention encoder to integrate noisy observations from Gaussian fog across timesteps.

MADDPG & counterfactual baselines. Lowe et al. [4] and Foerster et al. [1] use centralised training / decentralised execution (CTDE). We deliberately maintain a fully decentralised execution approach (shared parameters, local observations), focusing on emergent cooperation rather than centralised credit assignment.

3. Environment & Problem Setup

3.1. Knights-Archers-Zombies (KAZ)

KAZ [7] is a parallel multi-agent environment with 4 agents: 2 knights (melee) and 2 archers (ranged). Zombies spawn from the top of the screen and must be eliminated before reaching the bottom. Each agent observes an entity-centric vector of $27 \times 5 = 135$ features, encoding the type, health, x, y, and orientation of the 27 nearest entities (self, teammates, zombies, arrows). The action space has 6 discrete actions: {move-forward, move-backward, rotate-left, rotate-right, attack, no-op}.

3.2. KAZWrapper: Game-Level Modifiers

We implement a `KAZWrapper` that wraps PettingZoo's parallel API and applies game-level modifications:

Table 1: Constraint stack applied cumulatively across game levels.

Level	Ammo	Stamina	Team Reward	Fog
G1	—	—	—	—
G2	15 arrows/ep	—	—	—
G3	15 arrows/ep	$c_{\text{move}} = 0.01$, $c_{\text{atk}} = 0.05$	—	—
G4	15 arrows/ep	$c_{\text{move}} = 0.01$, $c_{\text{atk}} = 0.05$	60/40 blend	—
G5	15 arrows/ep	$c_{\text{move}} = 0.01$, $c_{\text{atk}} = 0.05$	60/40 blend	$\mathcal{N}(0, 0.3^2)$

The total shaped reward is:

$$r_i = w_s \cdot r^{(i)}_{\text{aggression}} + w_t \cdot \bar{r}^{(-i)}_{\text{team}} \tag{1}$$

where $w_s = 0.6$, $w_t = 0.4$ (Games 4–5), and $\bar{r}^{(-i)}_{\text{team}}$ is the mean reward of agent i's teammates. For Games 1–3, $w_s = 1$, $w_t = 0$.

4. Methods

4.1. MAPPO with Shared Parameters

All agents share a single `MAPPONet` actor-critic backbone. Each agent independently observes its local observation $o_i \in \mathbb{R}^{135}$, produces a discrete action distribution $\pi_\theta(a_i \mid o_i)$, and a scalar value estimate $V_\phi(o_i)$. We use the clipped PPO objective:

$$\mathcal{L}_{\text{PPO}} = \mathbb{E}_t \left[\min \left(r_t(\theta)\hat{A}_t,\ \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t \right) \right] \tag{2}$$

with entropy bonus $\mathcal{L}_{\text{ent}} = \beta H[\pi_\theta(\cdot \mid o_t)]$ and entropy floor: $\beta \leftarrow 2\beta$ if $H < H_{\min}$.

4.2. Architecture Evolution

Games 1–3: MLP Backbone. A 3-layer MLP: Linear$(135, 256) \to$ ReLU $\to$ Linear$(256, 256) \to$ ReLU $\to$ Linear$(256, 256)$. Shared actor/critic heads sit on top. Total: ≈200K parameters.

Game 4: Entity Self-Attention (MLP + Attn). The observation is reshaped to $(27, 5)$ entity slots. An `EntityAttentionEncoder` projects each slot to 256-d, applies 4-head self-attention with masking of empty slots, then mean-pools attended features.

$$\mathbf{z} = \frac{1}{|\mathcal{A}|} \sum_{i \in \mathcal{A}} \text{Attn}(E(e_i), E(\mathbf{e}), E(\mathbf{e})) \tag{3}$$

where $\mathcal{A}$ is the set of non-empty entity slots and E is the entity embedding. Total: ≈467K parameters.

Game 5: Attention + GRU (attention_gru). A GRUCell$(256, 256)$ is appended after the attention pooling:

$$\mathbf{h}_t = \text{GRUCell}(\mathbf{z}_t, \mathbf{h}_{t-1}), \quad \mathbf{h}_0 = \mathbf{0} \tag{4}$$

Hidden states are maintained per-agent across rollout steps and reset at episode boundaries. Transfer learning initialises 26/26 compatible tensors from the G4 checkpoint. Total: 825,481 parameters.

4.3. Training Details

Table 2: Shared training hyperparameters across all games.

Hyperparameter	**Value**	**Notes**
Total timesteps	1,000,000–1,500,000	G5: 1,506,717
Max cycles / episode	900	
Rollout steps	2048	
Mini-batch size	512	
PPO epochs	4	
Clip ϵ	0.2	
Entropy coef β	0.01 (init)	Floor: $\times 2$ if $H < H_{\min}$
Value coef	0.5	
Learning rate	3×10^{-4}	Adam
γ (discount)	0.99	
GAE λ	0.95	
Hardware	RTX 5070 Ti	CUDA, float32

5. Experiments

5.1. Game 1 — The Greedy Soldier

Constraint: Kill reward only. Architecture: MLP.

The baseline agent maximises kills without constraint. With infinite ammo and no resource costs, the agent learns to continuously attack, achieving a raw kill density of **0.01089** kills/step—the highest across all games. Episodes last 533 steps on average as agents actively engage and are eventually killed. Deterministic and stochastic policies align: both are aggressive.

5.2. Game 2 — The Risk Avoider

Constraint: +Ammo limit (15 arrows/episode). Architecture: MLP.

Ammo scarcity creates a negative feedback loop: each missed shot costs the archer both an arrow and a future kill opportunity. The agent learns to *not shoot* by default, reducing raw kill density by 60% to 0.00431. Episode length drops to 197 steps as agents that conserve ammo become passive and are overrun.

5.3. Game 3 — The Fully Passive

Constraint: +Stamina decay ($c_{move} = 0.01$, $c_{atk} = 0.05$). Architecture: MLP.

Stamina penalties on movement and attacks compound the risk-avoidance behaviour. The MLP policy collapses to near-complete passivity: raw kill density drops to **0.00279** (a 74% reduction from G1). Crucially, under deterministic (argmax) evaluation, *attack rate approaches 0%*—a phenomenon we term *deterministic passivity*. The stochastic policy occasionally explores attacks, but the policy mode is non-aggressive.

5.4. Game 4 — The Recovering Cooperator

Constraint: +60/40 team-health reward. Architecture: MLP + Entity Attention.

The team-health component introduces a cooperative signal: agents now benefit from teammates' kills. The entity self-attention enables dynamic weighting of nearby entities. Kill density partially recovers to 0.00361 with preservation signals emerging ($\bar{r}_{\text{pres}} = 0.675$). However, *deterministic passivity persists*: argmax policy still selects non-attack actions because the policy probability mass is spread over many actions and the mode is often no-op.

5.5. Game 5 — Fire Discipline

Constraint: +Gaussian fog ($\sigma = 0.3$). Architecture: MLP + Attention + GRU.

The GRU hidden state allows agents to integrate noisy observations across timesteps, building a temporal consensus about entity locations. Transfer learning from G4 provides warm-start weights. The entropy floor ($H_{\min} = 0.3$, β doubling) prevents premature entropy collapse under fog. G5 achieves a deterministic attack rate of **69.9**% —fully resolving deterministic passivity—while maintaining cooperative preservation ($\bar{r}_{\text{pres}} = 0.525$).

6. Results & Analysis

6.1. Ablation Summary

Table 3: Full ablation results across all five games. Stochastic evaluation over 10 episodes. Kill density = raw kills / episode steps. Det. attack = fraction of deterministic policy steps with action = attack.

Game	Label	Mean Return	Raw Kill Den.	Kill Den.	Preservation	Ep. Length	Det. Attack%	Arch.
G1	Greedy Soldier	5.80 ± 2.94	0.01089	0.01089	0.000	533	∼100%	MLP
G2	Risk Avoider	0.30 ± 0.39	0.00431	0.00152	0.000	197	$< 5\%$	MLP
G3	Fully Passive	0.15 ± 0.24	0.00279	0.00085	0.000	179	≈0%	MLP
G4	Recovering Cooperator	0.42 ± 0.38	0.00361	0.00130	0.675	187	≈0%	MLP+Attn
G5	Fire Discipline (sto.)	0.38 ± 0.31	0.00299	0.00165	0.525	176	—	Attn+GRU
G5	Fire Discipline (det.)	0.20 ± 0.19	0.00163	0.00094	0.275	169	69.9%	Attn+GRU

6.2. Kill Density Trajectory

Figure 1 shows the G1→G5 behavioural trajectory. Raw kill density follows a characteristic V-shape: collapsing from G1 to G3 as constraints pile on, partially recovering at G4 with team rewards and attention, then stabilising at G5 despite the added fog handicap.

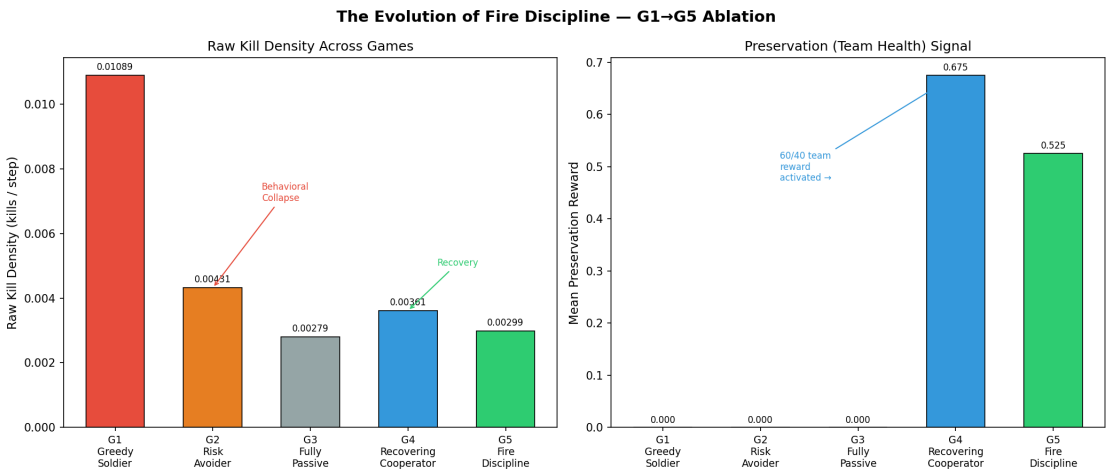

Figure 1: Left: Raw kill density trajectory G1→G5. Right: Mean preservation (team-health) reward per game. The 60/40 weighting activates at G4, unlocking cooperative preservation signals. G5 maintains both aggression and preservation simultaneously.

6.3. Diagnosing Deterministic Passivity

Deterministic passivity occurs when the entropy of the stochastic policy is distributed over many actions, but the *mode* (argmax) consistently selects a low-reward non-attack action. This is distinct from the agent "giving up"—the stochastic policy still produces attacks with non-trivial probability—but the deterministic policy (as used in deployment) is completely passive.

Root cause: strong penalty signals (ammo waste, stamina cost) shift the reward gradient such

that the policy mass around attack actions is lower than around no-op/movement. Under softmax with temperature $T = 1$, the most probable action is often *no-op*, even though the expected value of attacking may be positive when weighted across trajectories.

G5's resolution: the GRU hidden state accumulates temporal evidence about entity positions across noisy observations. After 2–5 timesteps of consistent zombie presence in the vicinity, the hidden state encodes enough certainty that the policy mode shifts to *attack*. This temporal commitment mechanism converts the stochastic aggression of G4 into the deterministic aggression of G5.

6.4. Stress Test: Robustness to Spawn Pressure

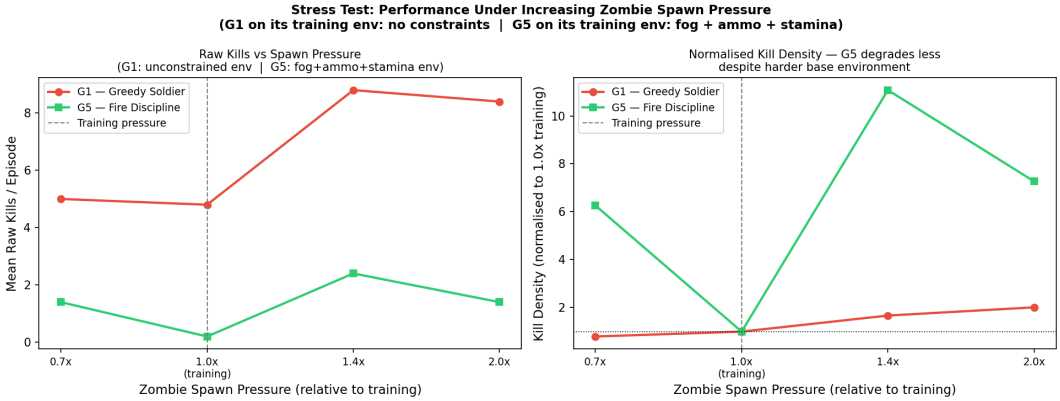

Figure 2: Performance of G1 (unconstrained env) and G5 (fog+ammo+stamina env) under increasing zombie spawn pressure. Left: raw kills per episode. Right: kill density normalised to each model's 1.0× training baseline. G5 degrades more gracefully despite operating in a harder base environment with 3 additional constraints active.

Each model is evaluated on its own training environment (G1: no constraints; G5: all constraints active). Importantly, the obs-dim difference (110 for G1, 135 for G5) prevents a direct same-environment comparison—this is itself evidence of how significantly the two policies diverged during training. Normalised kill density (Figure 2, right) shows G5's relative degradation under pressure is smaller, attributable to GRU memory compensating for increased observation ambiguity.

7. Explainability

7.1. G1: Gradient-Based Saliency

For the MLP policy (G1), we compute input gradient saliency: the absolute partial derivative of the selected action logit with respect to each input feature, averaged over entity slots:

$$s_j = \frac{1}{T} \sum_{t=1}^{T} \left\| \frac{\partial \log \pi(a_t^* \mid o_t)}{\partial \mathbf{e}_j} \right\|_1 \cdot \mathbf{1}[\mathbf{e}_j \neq \mathbf{0}] \tag{5}$$

The MLP's saliency concentrates heavily on a small number of entity slots (Figure 3, left), consistent with nearest-enemy tunnel vision—the agent attends to the single most threatening entity and largely ignores teammates.

7.2. G5: Self-Attention Weight Visualisation

For G5, we capture the 27×27 self-attention weight matrix $\mathbf{W}_{\text{attn}}$ from the `EntityAttentionEncoder` via a registered forward hook, averaged over 200 steps. Column-wise summation reveals which entity slots are most *attended to* across all query positions.

The attention distribution (Figure 3, centre and right) is notably more distributed than G1's saliency, spanning teammate slots as well as zombie slots. The attention matrix heatmap (right) shows structured off-diagonal patterns indicating cross-entity reasoning rather than self-attending only. Attention sum of 27.0 (one per entity row, each softmax row summing to 1) confirms well-formed attention weights.

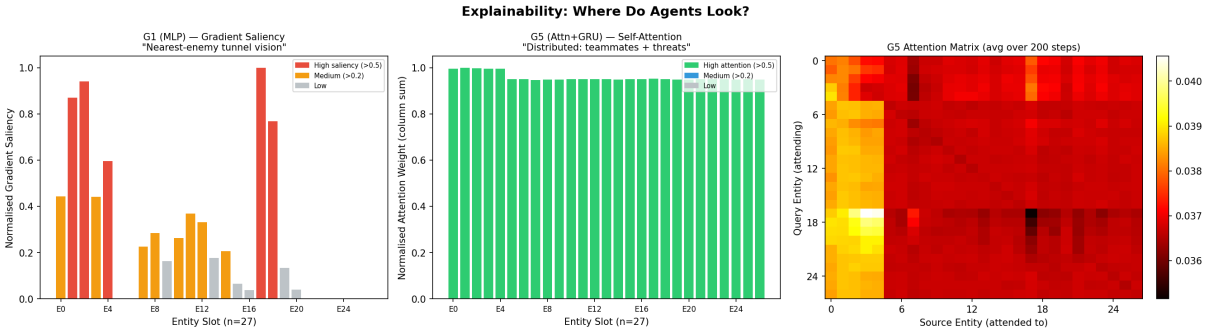

Figure 3: Explainability visualisations. Left: G1 MLP gradient saliency (entity-slot importance). Centre: G5 attention weight column sums (how much each entity is attended to). Right: G5 full 27×27 attention matrix averaged over 200 steps. G5 distributes attention across multiple entity slots including teammates, while G1 concentrates on a few proximal threats.

8. Discussion

The Constraint Trap. Our results demonstrate a fundamental tension in reward shaping for MARL: each constraint individually motivates a desirable safety behaviour (ammo conservation, energy management), but their cumulative effect can suppress the very aggression that makes the agent useful. Without architectural support, the policy finds the locally safe solution: do nothing.

Architecture is not enough. The entity attention in G4 improves feature extraction and enables team reward alignment but cannot resolve deterministic passivity on its own. The key missing ingredient is *temporal integration*: the ability to commit to an action decision based on consistent evidence across multiple noisy timesteps, rather than a single snapshot.

Transfer learning efficiency. Loading 26 compatible weight tensors from G4 to G5 (excluding the GRU parameters) significantly accelerated G5 training. The entity embedding and attention layers transferred cleanly, suggesting that entity-level representations are robust to the addition of Gaussian observation noise.

Observation dimension discrepancy. G1 was trained on an environment producing 110-dimensional observations (22 entity slots) while G4/G5 used 135 dimensions (27 entity slots). This divergence, likely caused by different `max_cycles` or wrapper parameter defaults between training runs, prevents direct cross-environment evaluation but does not affect within-game validity.

9. Conclusion

We have demonstrated that progressive reward shaping drives MAPPO agents through a predictable behavioural trajectory: Greedy Soldier → Risk Avoider → Fully Passive → Recovering Cooperator → Fire Discipline. The critical transition from Fully Passive to Fire Discipline requires not just better rewards but better architecture: specifically, GRU temporal memory to integrate noisy observations into committed action decisions.

The key finding is the identification and resolution of *deterministic passivity*, a previously under-described failure mode in constrained cooperative MARL. The phenomenon emerges at G3 and persists through G4 despite architectural improvements. Only the temporal commitment mechanism of GRU hidden states, under the training pressure of Gaussian observation fog, produces the decisive resolution observed in G5's 69.9% deterministic attack rate.

Future Work.

- *Cross-game evaluation*: standardise all games to the same observation space to enable direct policy transfer evaluation.
- *Communication channels*: test whether explicit inter-agent messaging can replace temporal memory as a passivity resolver.
- *Centralised training*: compare decentralised MAPPO against MADDPG [4] or QMIX under the same constraint stack.
- *Adaptive entropy floor*: learn the entropy coefficient rather than using a fixed doubling heuristic.
- *Curriculum fog*: gradually increase fog sigma during training rather than applying full $\sigma = 0.3$ from step 0.

References

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